

Here, h_ℓ^a is the hidden state from the previous attention layer, w_{norm} is a scaling vector, and s_ℓ^a is a normalization factor.

The nonlinear point-wise function g allows neural networks to learn complex transformations that cannot be represented by linear transformations. Here, we refer to the input dimensions of g as neurons. We can break down the MLP output m_ℓ into a sum over individual neuron contributions:

$$m_\ell = \sum_{i=1}^{\text{\#neurons}} \text{diag}(W_{\text{down}}^i) \cdot g^i \cdot \text{diag}(W_{\text{up}}^i) \cdot \text{diag}(w_{\text{norm}}) \cdot \frac{h_\ell^a}{s_\ell^a}$$

To compute the contribution of the i -th neuron in MLP layer ℓ to the logit output, we can treat g and s_ℓ^a as constants and use following projection matrix:

$$P_\ell^{m_i} = P \cdot \text{diag}(W_{\text{down}}^i) \cdot (g^i I) \cdot \text{diag}(W_{\text{up}}^i) \cdot \text{diag}(w_{\text{norm}}) \cdot \left((s_\ell^a)^{-1} I \right)$$

Using this projection matrix, we can pinpoint how much a single MLP neuron contributes to the model’s final output logits. Additionally, the same projection matrix allows us to trace how residual vectors from earlier layers influence the final output logits through that specific neuron via the residual stream decomposition.

3 Examples

In this section, we apply the *prisms* proposed earlier to explore how the **gemma-2b** model works internally in two examples. We use the **gemma-2b** model because it’s small enough to run on a standard PC without a dedicated GPU.

Retrieving capital city. First, let’s see how the model retrieves factual information to answer this simple question:

The capital city of France is ___

The model correctly predicts **Paris** as the most likely next token. To understand how it arrives at this prediction, we’ll use the prisms from the previous section.

We start by using the residual prism to plot how much each layer contributes to the logit output for several candidate tokens (different capital cities). Comparing the prediction logit of the right answer to reasonable alternatives can reveal important information about the network’s decision process.

Figure 2 shows each layer’s logit contribution for multiple candidate tokens. Some strong signals stand out, with large positive and negative contributions in the first and last layers. These likely mean these layers play key roles in the model’s predictions. Interestingly, there’s a strong positive contribution at the start, followed by an equally strong negative contribution in the next layer (the

first attention output). This might be because the **gemma-2b** model’s embedding and unembedding vectors are the same. So the input token strongly predicts itself as the output (due to the nature of the dot product operation). The network has to balance this out with a strong negative contribution in the next layer.

Figure 2 B zooms in to compare logit contributions at each layer for different targets. The a_{15} contribution stands out between **Paris** and other candidates. At this layer, the attention output aligns much more with the unembedding vector of **Paris** than other candidates.

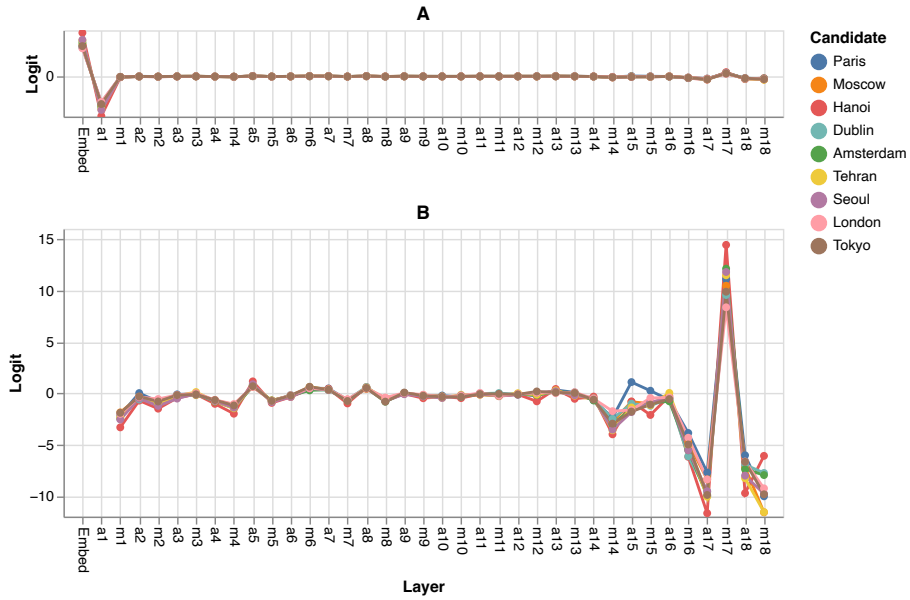


Figure 2: Logit contribution of each layer for different target tokens. Figure A shows the contributions of all layers, while Figure B zooms in on the contribution of the last layers.

We think a_{15} reads the correct output value from somewhere else via the attention mechanism, so we use the attention prism to decompose a_{15} into smaller pieces. Figure 3 shows how much each input token influences the output logits via the attention layer 15. The **France** token heavily affects the output through attention head 6 of the layer, which makes very much sense as **France** should somehow inform the network to output the correct capital city.

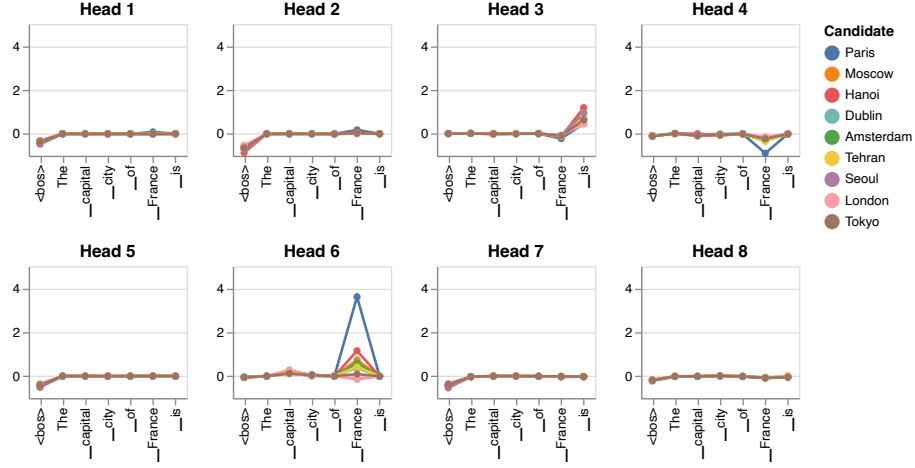


Figure 3: Logit contribution of each input token through attention heads at layer 15.

Next, we again use the residual prism to decompose the attention head 6 logits into smaller pieces. Figure 4 shows how the residual outputs from all previous layers at the **France** token contribute to the output logit via attention head 6. Interestingly, the **France** embedding vector contributes the most to the output logit. This indicates that the embedding vector of **France** somehow already includes the information about its capital city, and this information can be read easily by attention head 6.

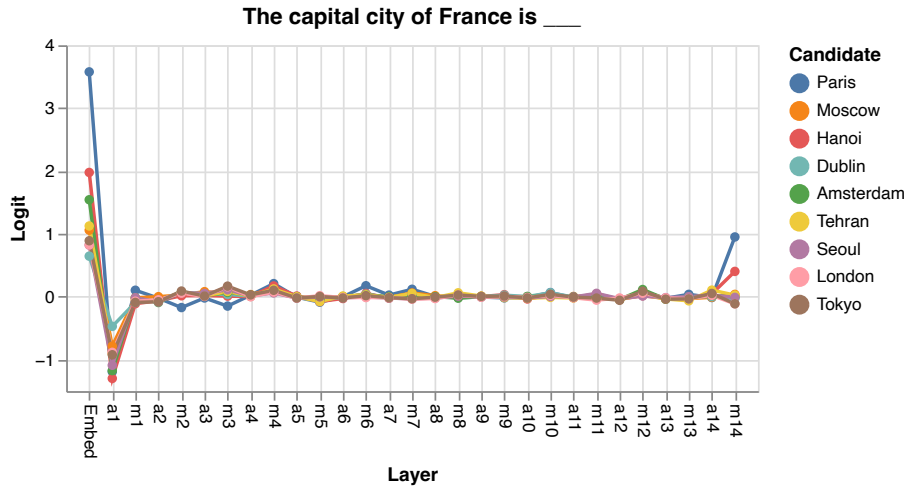


Figure 4: Logit contribution of all residual outputs through attention head 6 at layer 15.